
Table of Contents

.....	1
===== PARAMETERS =====	1
===== TRANSMITTER =====	1
===== CYCLIC PREFIX =====	2
===== MULTIPATH CHANNEL =====	2
===== SNR LOOP =====	2
Pass signal through multipath channel	2
Add AWGN noise	2
Remove cyclic prefix	3
FFT	3
Channel Equalization	3
Convert to serial	3
16-QAM Demodulation	3
BER Calculation	3
===== BER PLOT =====	3
===== CONSTELLATION =====	4

```
clc;
clear;
close all;
```

===== PARAMETERS

=====

```
M = 16;                % 16-QAM
k = log2(M);

Nfft = 64;             % Number of OFDM subcarriers
Nofdm = 100;           % Number of OFDM symbols
CP = 16;               % Cyclic Prefix length

SNR_dB = 0:5:25;       % SNR values
BER = zeros(size(SNR_dB));
```

===== TRANSMITTER

=====

```
% Generate random bits
Nbits = Nofdm * Nfft * k;
txBits = randi([0 1], Nbits, 1);

% 16-QAM modulation
txSymbols = qammod(txBits, M, ...
    'InputType', 'bit', ...
    'UnitAveragePower', true);
```

```
% Arrange QAM symbols into OFDM symbols
txSymbolsMatrix = reshape(txSymbols, Nfft, Nofdm);
```

```
% IFFT
txOFDM = ifft(txSymbolsMatrix, Nfft);
```

===== CYCLIC PREFIX

=====

```
% Copy last CP samples to the beginning
cyclicPrefix = txOFDM(end-CP+1:end, :);
```

```
% Add cyclic prefix
txOFDM_CP = [cyclicPrefix; txOFDM];
```

===== MULTIPATH CHANNEL

=====

```
% Multipath channel
h = [1 0.5 0.3];
```

```
% Channel frequency response
H = fft([h zeros(1, Nfft-length(h))], Nfft).';
```

===== SNR LOOP

=====

```
for i = 1:length(SNR_dB)
```

Pass signal through multipath channel

```
    rxOFDM_CP = zeros(size(txOFDM_CP));

    for n = 1:Nofdm
        rxOFDM_CP(:,n) = filter(h, 1, txOFDM_CP(:,n));
    end
```

Add AWGN noise

```
    rxOFDM_CP = awgn(rxOFDM_CP, SNR_dB(i), 'measured');
```

Remove cyclic prefix

```
rxOFDM = rxOFDM_CP(CP+1:end, :);
```

FFT

```
rxSymbolsMatrix = fft(rxOFDM, Nfft);
```

Channel Equalization

```
rxSymbolsMatrix = rxSymbolsMatrix ./ H;
```

Convert to serial

```
rxSymbols = rxSymbolsMatrix(:);
```

16-QAM Demodulation

```
rxBits = qamdemod(rxSymbols, M, ...  
    'OutputType', 'bit', ...  
    'UnitAveragePower', true);
```

BER Calculation

```
[numErrors, BER(i)] = biterr(txBits, rxBits);
```

```
fprintf('SNR = %2d dB | Errors = %d | BER = %e\n', ...  
    SNR_dB(i), numErrors, BER(i));
```

```
SNR = 0 dB | Errors = 8014 | BER = 3.130469e-01
```

```
SNR = 5 dB | Errors = 4943 | BER = 1.930859e-01
```

```
SNR = 10 dB | Errors = 2352 | BER = 9.187500e-02
```

```
SNR = 15 dB | Errors = 642 | BER = 2.507812e-02
```

```
SNR = 20 dB | Errors = 34 | BER = 1.328125e-03
```

```
SNR = 25 dB | Errors = 0 | BER = 0.000000e+00
```

```
end
```

===== BER PLOT

```
=====
```

```
figure;
```

```
semilogy(SNR_dB, BER, '-o', ...
```

```

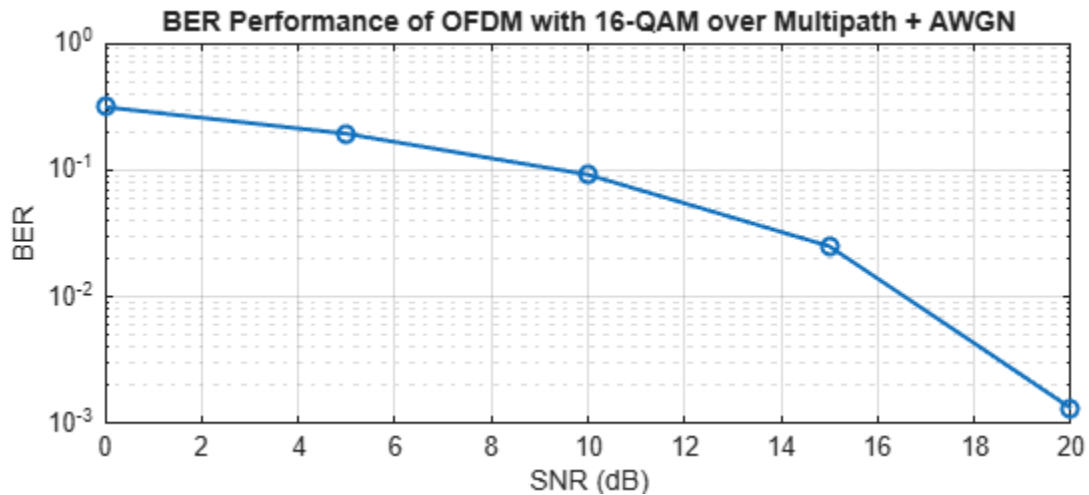
        'LineWidth', 1.5);

grid on;

xlabel('SNR (dB)');
ylabel('BER');

title('BER Performance of OFDM with 16-QAM over Multipath + AWGN');

```



===== CONSTELLATION

=====

```

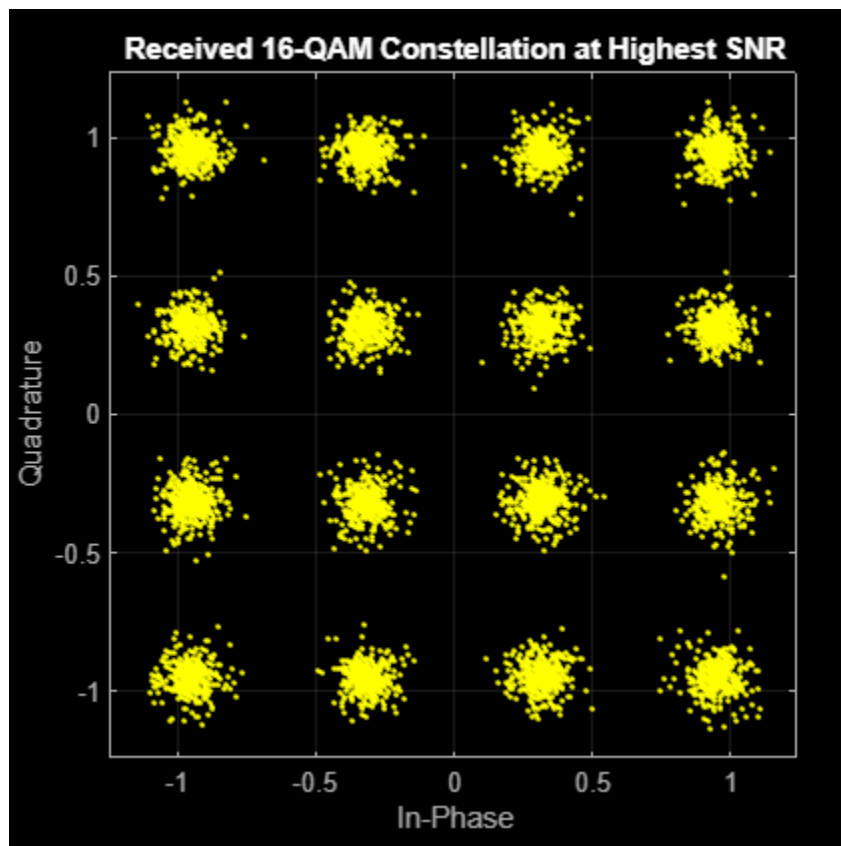
% Show constellation at highest SNR
figure;

scatterplot(rxSymbols);

title('Received 16-QAM Constellation at Highest SNR');

grid on;

```



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